Analysis of Shunt LDO With Physical Security to Power/EM Side-Channel Attacks

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***Abstract*—With growing concerns over physical security against power and electromagnetic side-channel attacks (SCA), this work provides a comprehensive analysis of different existing isolation countermeasures, especially the analysis of security-focused shunt low-dropout regulators (LDOs). Using both small-signal and large-signal analysis, we demonstrate that shunt LDOs offer superior security using the metric current-domain signature attenuation (CDSA) and load-to-supply isolation (LSI) performance while mitigating mid-frequency power-supply ripple-rejection (PSR) peaking. Various existing shunt LDO architectures are evaluated, highlighting trade-offs between DC rejection and high-frequency CDSA response. Alternative isolation techniques such as galvanic isolation, digital LDOs, and integrated voltage regulators (IVRs) are also examined.**

***Index Terms*—Series LDO, shunt LDO, power supply ripple rejection (PSR), power side-channel attack/analysis (SCA), security, current domain signature attenuation (CDSA), load-to-supply isolation (LSI).**

I. Introduction

THE switching of digital circuits, specifically during the transition of the clock edge, can induce noise on the main supply and leak encrypted information, especially since encryption algorithms are typically implemented using digital circuits. Despite the mathematical security of cryptographic algorithms, many researchers have shown that keys can be decrypted through physical side-channel methods such as electromagnetic (EM) emissions, supply power, timing, etc. [1], [2]. Other mixed-signal circuits, such as flash and SAR ADCs, have also been shown to be vulnerable to attack through side-channel methods [3], [4]. Power SCA detects the switching current of a digital crypto core through an external sensing resistor at the supply and runs correlation algorithms with the secret keys for decryption [5], [6]. SCA can be mitigated through isolation [7], [8], [9], logic-level masking [10], [11], [12], [13], or the injection of noise through various circuits such as LDOs or switched-capacitor banks [14], [15], [16], [17]. In practice, EM SCA is more prevalent as a non-invasive attack. Some EM SCA countermeasures include different layout techniques and the addition of decoupling capacitors [18], [19], [20]. EM SCA relies on a near-field H-probe to retrieve the di/dt information when transistors switch during clock transitions. This leakage is present not only within the circuits, but also through the pads and wire bonding. Therefore, effective isolation from pads and wire bonding can help reduce the EM near field and become more resilient to EM SCA.

This paper focuses on the inherent security benefits of using the LDO as an isolation element without any noise injection; thus, it involves another important security metric, current-domain signature attenuation (CDSA), which was introduced in [21] and [22]. The attenuation is defined from the load-current ripple to the supply-current ripple and is given by the equation

$$CDSA = i_l/i_{in} = \frac{1}{A_L}, \tag{1}$$

where i_l is the load current ripple, i_{in} is the supply current ripple, and A_L is the transfer function.

When analog and digital circuits are integrated on the same board, additional switching noise can be directed from the digital circuits to the more noise-sensitive analog circuits. The LDO for the analog circuits focuses on noise rejection from the main supply to the analog load supply [26], [27]. This metric is referred to as power-supply ripple rejection (PSR). The PSR equation is given by

$$PSR = v_{in}/v_{out} = \frac{1}{A_{IN}}, \tag{2}$$

where v_{in} is the input supply to the LDO, v_{out} is the output of the LDO, and A_{IN} is the transfer function from the input supply to the output load.

For digital circuits, the LDO should also focus on noise rejection from the digital load to the main supply. To quantify the ability of the LDO to suppress back-fed noise from the switching load to the supply, we include another LDO metric, load-to-supply isolation (LSI), the ratio of load-current ripple to another supply's voltage ripple, given by the equation

$$LSI = i_l/v_{vdda} = \frac{1}{A_{LSI}} = k \cdot CDSA, \tag{3}$$

where i_l is the digital switching load, v_{vdda} is the other connected analog supply, A_{LSI} is the transfer function, and k is the constant that relates CDSA and LSI, which will be explained further in Sec. II-D.

A. Shunt LDO Background

LDOs have been widely researched, including output-compensated LDOs and gate-compensated LDOs, along

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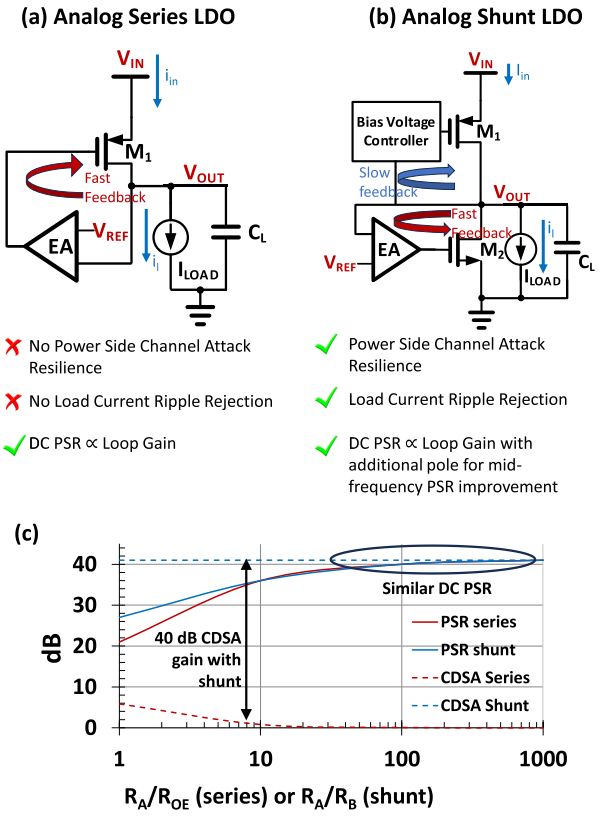


Fig. 1. (a) Sample analog series LDO. (b) Sample analog shunt LDO. (c) Series and Shunt LDO CDSA and PSR Comparison. R_A/R_{OE} and R_A/R_B are the output resistances of the error amplifier for series LDO or the current mirror bias for shunt LDO at the gate of the driving PMOS (this will be discussed further in Sec. II-A).

with various compensation methods to improve different LDO metrics [26], [28], [29].

This paper categorizes LDOs into two types: series LDOs and shunt LDOs. Typical output-compensated or gate-compensated LDOs are a type of series LDO with a feedback loop to the driving PMOS, with similar currents passing through both the driving PMOS and the load. Fig. 1(a) shows a typical output-compensated LDO. Fig. 1(b) shows a more security-focused shunt LDO. The security-focused shunt LDO has two feedback loops: a slow feedback to the driving PMOS and a fast feedback to the shunt path. Note the fundamental difference between the flipped voltage follower LDO and the security-focused LDO. This configuration allows the security-focused shunt LDO to decorrelate the load current from the input current, providing power SCA resilience and high backfeed noise rejection. Fig. 1(c) illustrates the benefit of the 40 dB CDSA proportional to the loop gain for the shunt LDO, while maintaining a similar DC PSR performance as the series LDO. Note that if the security focus is solely on high-frequency transitions, the shunt path can be reduced to a simple DC current bypass with slow feedback, while high-frequency compensation is provided by the load capacitor.

The evolution of security-focused shunt LDO integrated circuits is shown in Fig. 2 [31]. Security-focused shunt LDOs started with driving PMOS current sources (CS) with a high-loop-gain analog shunt path, as shown in Fig. 2(a) [7]. Digital control engages to adjust the number of PMOS slices that are on until the regulated voltage falls within the accepted range, and then disengages while the regulated voltage remains within the range; this is called switch-mode control (SMC) [32]. SMC can control the bandwidth (BW) of the driving PMOS feedback by tuning the threshold voltages, and when SMC disengages, the PMOS slices become constant current sources. In crypto engines, key-dependent leakages result in load current variations and can potentially trigger SMC changes. If the SMC threshold is invariant to key-dependent leakages, assuming no channel-length modulation, the load current variations will not be leaked through the supply, and the keys will remain concealed. Since sensitive information is leaked during clock transitions at mid-to-high frequency (>100 MHz), the analog shunt path consumes high power while having limited BW; thus, such a shunt path may provide only limited benefits. Fig. 2(a) shows 1M correlational power attack (CPA) minimum traces to disclosure (MTD) with a load capacitance of 450 pF.

To advance the shunt LDO with additional security benefits, the CS was changed to a cascoded PMOS configuration, while the shunt path was reduced to common-gate (CG) slices digitally controlled to connect to a bias voltage, as shown in Fig. 2(b) [21], [22]. The common-gate slices act as a local negative feedback (LNFB) to drain excess current from the quantized CS while providing some negative feedback. Note that LNFB is what the authors of these papers refer to as the shunt path without error amplifier. The bias voltages are used to bias the MOSFETs in the saturation region for both high output resistance and intrinsic gain. This configuration improved the MTD to 1 B with a load capacitance of 150 pF.

Fig. 2(c) introduces a digitally friendly bias for the CS with a ring oscillator (RO) that acts as both an LNFB and control for the CS [8], [23]. Even though the LDO and randomized RO configuration only achieves 390M MTD with a fixed load capacitance of 180 pF, the configuration with the LDO, randomized RO, and load capacitors achieves 1.25B MTD with sixteen approximately 20 pF variable load capacitors. The randomization of load capacitors constitutes a time-varying transfer function (TVTF).

Fig. 2(d) improves the previous bias with a digitally synthesizable dual bias for the cascoded PMOS CS [24], [25]. The MTD for this method is >200M with a load capacitance of 150 pF. These works on the security-focused shunt LDO have been shown to improve CDSA; however, detailed theory, analysis, and additional benefits of shunt LDOs have not been presented. This work aims to fill the gaps in the analysis of shunt LDOs; further analysis of the existing LDOs is explored in Sec. IV.

B. Other Isolation Methods for SCA

Fig. 3 presents three additional categories of isolation methods for SCA: switched capacitor converters, other LDO architectures, and integrated voltage regulators (IVR). Note that isolation methods do not need to function standalone but are best used in conjunction with other methods, such as noise injection or logic masking, for enhanced security. Forms of the switched capacitor converters in [9] and [33] present isolation methods by disconnecting power and/or ground while operating with a precharged capacitor during crypto-core switching, with one example shown in Fig. 3(a). [16], [34] utilizes an all-digital LDO (or digital series LDO) with noise injection at the gate of the driving PMOS, with one example shown in Fig. 3(b). Fig. 3(c) employs an IVR as isolation along with

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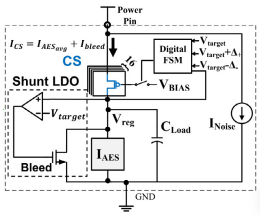
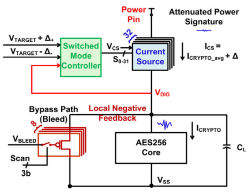
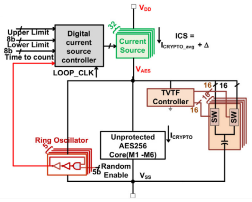
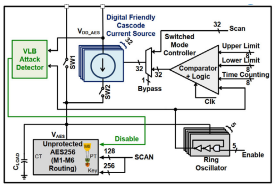
	Recent Shunt-LDO for Side-Channel Attack			
	(a) TCAS-I (2018) Das et al.	(b) ISSCC (2020) JSSC (2021) Das et al.	(c) ISSCC (2021) JSSC (2022) Ghosh et al.	(d) CICC (2022) OJ-SSCS (2025) Ghosh et al.
System Diagrams				
CPA MTD	>1M (1000x)	1B (>125000x)	w/ LDO & randomized RO: 390M Total: 1.25 B (>178000x)	w/ LDO only: >200M (10x)
Power Penalty	68%	50%	50%	50%
Area Penalty	60%	37%	52%	35%

Fig. 2. System diagrams, correlational power attack (CPA) minimum trace to disclosure (MTD), and specific implementation comments for previous shunt LDO ICs in physical security domain for power/EM SCA. (a) Analog shunt LDO with high error amplifier gain [7] (b) Cascoded driving PMOS with common gate shunt path [21], [22] (c) Digitally synthesizable current source with ring oscillator shunt path [8], [23] (d) Digitally synthesizable cascoded driving PMOS with ring oscillator shunt path [24], [25].

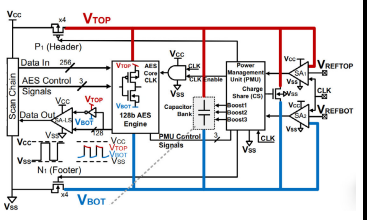
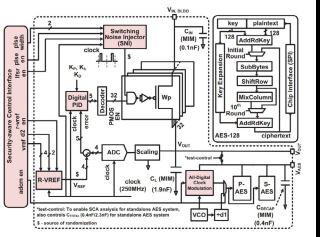
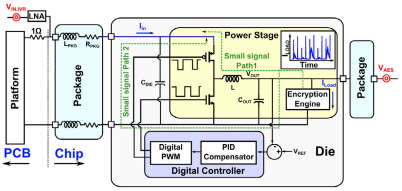
	Other Isolation Methods for Side-Channel Attack		
	(a) CICC (2021) Wang et al.	(b) ISSCC (2019) Singh et al.	(c) ISSCC (2017) Kar et al.
System Diagrams			
CPA MTD	>3M (600x)	6.8M (>3579x)	>100k (>20x)
Power Penalty	130%	32%	5%
Area Penalty	42.5%	36.9%	--

Fig. 3. Other isolation methods for SCA. (a) Galvanically isolated with integrated charge pump [9]. (b) All digital dropout regulator [16]. (c) Integrated voltage regulator (IVR) [30].

loop randomization for noise injection [30]. Further analysis can be found in Sec. V.

C. MTD Vs. SNR for Power SCA

As this paper presents mostly theoretical analysis of different isolation circuits in terms of CDSA, connecting CDSA back with standard SCA metric MTD becomes important. Many papers have shown that the number of measurements required for a successful attack (or MTD in our case) increases as the signal-to-noise ratio (SNR) of the measurements decreases [35], [36], [37]. $SNR = \frac{VAR(Q)}{VAR(N)}$, where Q is the measured power consumption, and N is the noise. Assuming that the isolation with the shunt LDO reduces the signal by a factor of c (or CDSA) and that noise is dominated by external sources such as equipment or environmental noise, then

$$SNR = \frac{VAR(Q)}{c^2 \cdot VAR(N)}. \tag{4}$$

By reducing the SNR, MTD improves. For the case of isolation in power SCA, c corresponds to CDSA; whereas in EM SCA, c takes into account the reduction due to pad and wire-bonding emanations.

Following the procedure in [35], assuming 1. a maximum correlation between the measured Q and the hypothetical power consumption of the correct key H, and 2. no timing misalignment, the maximum correlation between Q+N and H becomes

$$\rho_{max} = \frac{1}{\sqrt{\left(1 + \frac{1}{SNR}\right)}}. \tag{5}$$

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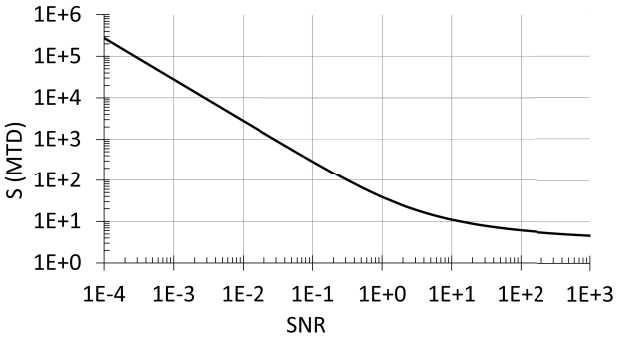


Fig. 4. MTD number vs. SNR using eq. 6 with $\alpha = 0.9999$ and $Z_\alpha = 3.719$ in [35].

Then, the number of traces S required to perform a differential power analysis (DPA) is

$$S = 3 + 8 \cdot \left(\frac{Z_\alpha}{\ln\left(\frac{1+\rho_{max}}{1-\rho_{max}}\right)} \right)^2, \tag{6}$$

where Z_α is the inverse standard normal CDF at α. [35] suggests $\alpha = 0.9999$ for high confidence in key revelation, which gives $Z_\alpha = 3.719$. Fig. 4 shows the corresponding plot for MTD vs. SNR using eq. 6. When SNR>1, reducing SNR does not significantly improve MTD; however, when SNR$\ll$ 1, MTD is linear in log-log scale, corresponding to a power-law decay where the trendline can be modeled as $MTD \approx 28.623 \cdot SNR^{-0.995}$. Therefore, when SNR$\ll$ 1, MTD is approximately inversely proportional to SNR and proportional to CDSA^2 without any noise injection.

D. Application and Contribution

This paper analyzes the benefits of security-focused LDOs in the following applications: 1. LDOs for circuits that require security or power/EM SCA resilience with high CDSA; 2. LDOs for digital circuits to provide high LSI that isolates digital switching noise from more sensitive analog circuit supplies; 3. LDOs for sensitive circuits to provide high PSR at mid-to-high frequencies. The contributions of this paper are organized as follows:

- A theoretical analysis comparing different existing isolation countermeasures for SCA using the CDSA metric. The paper focuses primarily on the analysis of security-focused shunt LDO architectures.
- Exploration of the potential of security-focused shunt LDOs as replacements for conventional power management LDOs, evaluated through PSR, CDSA, and LSI metrics. Shunt LDOs are shown to improve mid-frequency PSR, CDSA, and LSI responses with a simple compensation method.
- Special considerations for shunt LDOs and comparisons between existing analog and digital shunt LDOs are also presented. Analysis of existing shunt LDOs shows that low-frequency CDSA can be traded off for high-frequency CDSA to achieve a better response around clock rise or fall transitions, thereby improving power/EM SCA resilience.

The rest of the paper is organized as follows. Sec. II provides a theoretical analysis of the frequency responses of PSR, CDSA, and LSI for series and shunt LDOs. Sec. III describes additional design considerations for shunt LDOs and the use of digital LDOs. Sec. IV provides a complete comparison of both PSR and CDSA for existing shunt LDOs used to improve security. Sec. V analyzes additional existing isolation architectures in terms of CDSA security. Sec. VI concludes the paper.

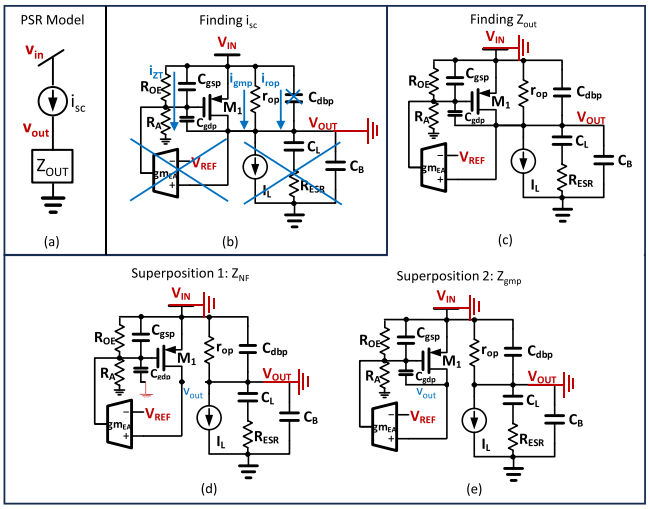


Fig. 5. Sample PSR analysis to find i_{sc} and Z_{out} in series LDO.

II. PSR, CDSA and LSI Analysis for Analog Series and Shunt LDOs

To visualize the benefits of shunt LDOs on PSR and CDSA, this section details a frequency domain analysis with the specific locations of the poles and zeros to further examine possibilities to improve the response. Theoretical analysis helps to pinpoint the potential areas of improvement in the existing architectures. Later in Sec. II-E, CDSA analysis is expanded into the LSI analysis to show the direct correlation.

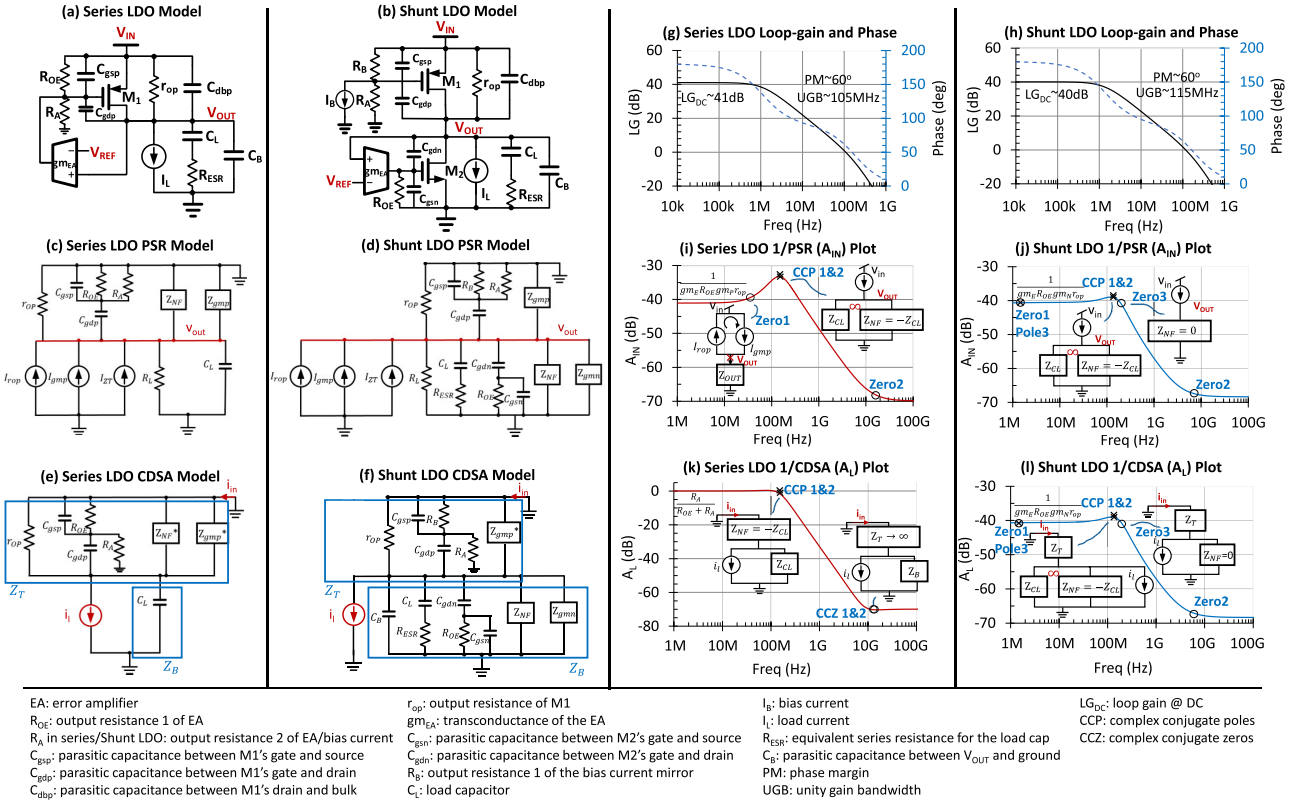
A. Assumptions and Conditions

To improve understanding, this paper starts by analyzing the PSR and CDSA for output-compensated analog LDOs with an on-chip load capacitor (C_L = 150 pF). PSR is the inverse of the transfer function A_{IN} from the supply to the output voltage (eq. 2). CDSA is the inverse of the transfer function A_L from the load current to the supply current (eq. 1). For the analysis, a relatively low-power application (i_L = 500 μA, V_{IN} = 1.2 V, and V_{OUT} = 1 V) is considered, with the driving PMOS always operating in the saturation region.

The series and shunt LDO models with parasitics are shown in Fig. 6(a) and (b). To simplify the analysis, the effects of C_{dbp}, C_B, and R_{ESR} are assumed to occur only at high frequencies and are therefore ignored in the plots and equations. Since the bias control to M1 operates at a much lower bandwidth than the feedback shunt loop (typically utilizing SMC), the bias voltage control to M1 in the shunt LDO has been replaced with a current mirror model for simplicity. The error amplifier (EA) has been modeled as a single-pole amplifier with a transconductance block and an output pole.

For a fair comparison, the series and shunt LDOs utilize different EA gains to obtain similar loop-gain (LG) responses, as shown in Fig. 6(g)-(h). Both LDOs achieve a DC LG of around 40 dB, a unity-gain bandwidth (UGB) of around 100 MHz, and a phase margin (PM) of 60 degrees. The series

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EA: error amplifier
R_{OE}: output resistance 1 of EA
R_A In series/Shunt LDO: output resistance 2 of EA/bias current
C_{gsp}: parasitic capacitance between M1's gate and source
C_{gdp}: parasitic capacitance between M1's gate and drain
C_{dbp}: parasitic capacitance between M1's drain and bulk
r_{op}: output resistance of M1
gm_{EA}: transconductance of the EA
C_{gsn}: parasitic capacitance between M2's gate and source
C_{gdn}: parasitic capacitance between M2's gate and drain
R_B: output resistance 1 of the bias current mirror
C_L: load capacitor
I_B: bias current
I_L: load current
R_{ESR}: equivalent series resistance for the load cap
C_B: parasitic capacitance between V_{OUT} and ground
PM: phase margin
UGB: unity gain bandwidth
LG_{DC}: loop gain @ DC
CCP: complex conjugate poles
CCZ: complex conjugate zeros

* CDSA model impedances are multiplied by an additional factor $Z^* = Z \cdot \frac{R_A + R_{OE} \| Z_{Cgsp}}{R_A}$, as explained in Sec. II-C1.

Fig. 6. (a)–(b) Analog series and shunt LDO models. (c)–(d) Series and shunt LDO PSR models with i_{sc} and Z_{out}. (e)–(f) Series and shunt LDO CDSA models with Z_T and Z_B. (g)–(h) Series and shunt LDO closed-loop gain and phase plots. (i)–(j) Series and shunt LDO A_{IN} plots. (k)–(l) Series and shunt LDO A_L plots. The plots are simulated with the following parameters in 65-nm technology: $AV_{EAseries}$ = 15, $AV_{EAshunt}$ = 150, $R_{OEseries} = R_B$ = 10 $k\Omega$, R_A = 1000 · R_B, $R_{OEshunt}$ = 100 $k\Omega$, C_L = 150 pF, $C_{gspshunt}$ = 10 pF, I_L = 500 μA, $(W/L)_1$ = 500, $(W/L)_2$ = 50. The effects of C_{dbp}, C_B, and R_{ESR} are assumed to occur at high frequencies and are therefore ignored in the plots and equations.

LDO has an EA gain ($Av_{EAseries}$) of 15, while the shunt LDO has an EA gain ($Av_{EAshunt}$) of 150. This difference arises because M1 in the series LDO incorporates an additional common-source gain stage, while M2 in the shunt LDO does not provide much gain due to lower gm_n and R_{out} from the smaller shunt-path current and smaller M2.

Another important assumption is that the series LDO assumes supply current replication through the EA [26], [27]; thus, for the output resistances of the EA, $R_A \gg R_{OE}$. The shunt LDO assumes $R_A \gg R_B$ with current mirror bias. The reason for a high R_A/R_{OE} and R_A/R_B ratio is shown in Fig. 1(c). As the ratio increases, PSR for both types of LDOs improves by over 10 dB, since the variation in v_{in} is mostly transferred to the gate of the PMOS, effectively canceling the supply noise. When v_{in} is preserved through the resistive divider at v_g of M1 ($v_{sg} \simeq 0$), $i_{sg} = gm_p \cdot v_{sg} \simeq 0$, thus reducing the effect of supply noise on the load. The CDSA response for the shunt LDO remains relatively constant, as the shunt path is the primary source adjusting for load current changes. The CDSA for the series LDO deteriorates, but not as significantly as the improvement in PSR, because most of the load variation is still supplied by i_{in}.

B. PSR Analysis

PSR analysis in this section is performed using the short-circuit current method through the transfer function

$$A_{IN} = v_{out}/v_{in} = gm_{sc} \cdot Z_{\text{out}} \tag{7}$$

where $gm_{sc} = i_{sc}/v_{in}$, as shown in Fig. 5(a). Here, i_{sc} is the short-circuit current from the input when the output node is shorted, and Z_{out} is the output impedance when the input node is shorted. Therefore, when v_{in} = 1,

$$A_{IN} = v_{out} = i_{sc} \cdot Z_{out} \tag{8}$$

.

1) PSR Model Descriptions: The short-circuit currents (i_{sc}) are calculated by shorting the output node and computing i_{in}, as shown in Fig. 5(b). The three i_{sc} components given by Eq. 9 are: the current i_{rop} due to r_{op}; the current i_{gmp} due to the M1 transconductance (gm_p), caused by the voltage divider at the gate of M1; and the current i_{Z_T} due to the top parasitic impedance through C_{gdp}. Note that both series and shunt LDOs have similar i_{sc}, except that the shunt LDO has R_B instead of R_{OE} in the series LDO; these resistances occur at the gate of M1.

$$\begin{aligned} i_{rop} &= gm_{rop} \cdot v_{in} = \frac{1}{r_{op}} \cdot v_{in} \\ i_{gmp} &= gm_{gmp} \cdot v_{in} = \frac{gm_p(R_{OE} \| Z_{Cgsp})}{R_{OE} \| Z_{Cgsp} + R_A \| Z_{Cgdp}} \cdot v_{in} \\ i_{Z_T} &= gm_{Z_T} \cdot v_{in} = \frac{R_A/(R_A + Z_{Cgdp})}{R_{OE} \| Z_{Cgsp} + R_A \| Z_{Cgdp}} \cdot v_{in} \end{aligned} \tag{9}$$

Output impedance (Z_{out}) is calculated by shorting v_{in}, as shown in Fig. 5(c). The straightforward resistances and capacitances are directly shown in the model in Fig. 6(c)-(d); other

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TABLE I

POLE/ZERO EQUATIONS FOR A_{IN} AND A_L FOR ANALOG SERIES AND SHUNT LDOS

Assume $R_A \to \infty$	A_{IN} (1/PSR = v_{out}/v_{in})		A_L (1/CDSA = i_{in}/i_l)	
	Analog Series LDO	**Analog Shunt LDO**	**Analog Series LDO**	**Analog Shunt LDO**
DC	$\frac{R_L}{gm_{EAP}R_Lr_{op}+R_L+r_{op}} \approx \frac{1}{gm_{EAP}r_{op}}$	$\frac{R_L}{gm_{EAN}R_Lr_{op}+R_L+r_{op}} \approx \frac{1}{gm_{EAN}r_{op}}$	$\frac{R_A}{R_A+R_{OE}} \approx 1$	$\frac{1}{gm_{EAN}r_{op}}$
Zero 1	$I_{rop}+I_{gmp}=0$	$I_{rop}+I_{gmp}=0$	----	$r_{op}=-Z^*_{gmp}$
	$s=-\frac{1}{[R_{OE}(C_{gdp}(gm_pr_{op}+1)+C_{gsp})]}$	$s=-\frac{1}{[R_B(C_{gdp}(gm_pr_{op}+1)+C_{gsp})]}$	----	$s=-\frac{1}{[R_B(C_{gdp}(gm_pr_{op}+1)+C_{gsp})]}$
Zero 2*	$I_{rop}+I_{gmpHF}+I_{ZTHF}=0$	$I_{rop}+I_{gmpHF}+I_{ZTHF}=0$	$Z_T \to \infty$	$r_{op} \parallel (Z_{Cgdp}+Z_{Cgsp}) \parallel Z^*_{gmpHF} \to \infty$
	$s=-\frac{C_{gdp}(gm_pr_{op}+1)+C_{gsp}}{C_{gdp}C_{gsp}r_{op}}$	$s=-\frac{C_{gdp}(gm_pr_{op}+1)+C_{gsp}}{C_{gdp}C_{gsp}r_{op}}$	$r_{op} \parallel Z^*_{NF} \parallel Z^*_{gmp} \parallel (Z_{Cgdp}+R_{OE} \parallel Z_{Cgsp}) \to \infty$	$s=-\frac{C_{gdp}(gm_pr_{op}+1)+C_{gsp}}{C_{gdp}C_{gsp}r_{op}}$
Zero 3	----	$Z_{NF}=0$	$CCZ\ at\ \approx |s| = \frac{C_{gdp}(gm_pr_{op}+1)+C_{gsp}}{C_{gdp}C_{gsp}r_{op}}$	$Z_{NF}=0$
	----	$s=-\frac{1}{R_{OE}C_{gn}}$		$s=-\frac{1}{R_{OE}C_{gn}}$
Pole 1 & 2	$Z_{CL} \parallel Z_{NF} \to \infty$ or $Z_{CL}=-Z_{NF}$	$Z_{CL}=Z_{NF}$	$Z_{CL}=-Z_{NF}$	$Z_{CL}=-Z_{NF}$
	$s=-\frac{1}{2R_{OE}C_{gp}} \pm \frac{\sqrt{C_L(-4gm_{EAP}R_{OE}C_{gp}+C_L)}}{2C_LR_{OE}C_{gp}}$	$s=-\frac{1}{2R_{OE}C_{gn}} \pm \frac{\sqrt{C_L(-4gm_{EAN}R_{OE}C_{gn}+C_L)}}{2C_LR_{OE}C_{gn}}$	$s=-\frac{1}{2R_{OE}C_{gp}} \pm \frac{\sqrt{C_L(-4gm_{EAP}R_{OE}C_{gp}+C_L)}}{2C_LR_{OE}C_{gp}s}$	$s=-\frac{1}{2R_{OE}C_{gn}} \pm \frac{\sqrt{C_L(-4gm_{EAN}R_{OE}C_{gn}+C_L)}}{2C_LR_{OE}C_{gn}}$
Pole 3	----	$Z_{CL}=-Z_{gmp}$	----	$Z_{CL}=-Z_{gmp}$
	----	$s=-\frac{gm_pR_BC_{gdp}+C_L}{R_BC_L(C_{gdp}+C_{gsp})} \approx -\frac{1}{R_B\ (C_{gdp}+C_{gsp})}$	----	$s=-\frac{gm_pR_BC_{gdp}+C_L}{R_BC_L(C_{gdp}+C_{gsp})} \approx -\frac{1}{R_B\ (C_{gdp}+C_{gsp})}$
Peaking	$4A_{EA}gm_pR_{OE}C_{gp} > C_L$	$4A_{EA}gm_nR_{OE}C_{gn} > C_L$	*negligible as* $A_L \approx 1$ *already*	$4A_{EA}gm_nR_{OE}C_{gn} > C_L$

$gm_{EAP}=gm_{EA}R_{OE}\ gm_P$ $gm_{EAN}=gm_{EA}R_{OE}\ gm_n$ $C_{gn}=C_{gdn}+C_{gsn}$ $C_{gp}=C_{gdp}+C_{gsp}$ $I_{gmpHF}=\frac{gm_pZ_{Cgsp}}{Z_{Cgsp}+Z_{Cgdp}}$ $I_{ZTHF}=\frac{1}{Z_{Cgsp}+Z_{Cgdp}}$ $Z^*_{gmpHF}=\frac{Z_{Cgsp}+Z_{Cgdp}}{gm_pZ_{Cgsp}}$

*: Only an approximation. Only parasitic caps of Cgs and Cgd are considered, other high frequency effects of the transistors are not considered.

components involving different transconductances are given in Eq. 10 for the series LDO and Eq. 11 for the shunt LDO, using superposition as shown in Fig. 5(d) and (e). Z_{NF} is the negative-feedback impedance without the voltage divider at the gate of M1; Z_{gmp} is the impedance that primarily involves gm_p, caused by the voltage divider at the gate of M1, where the effect of Z_{NF} is subtracted; Z_{gmn} is the impedance that involves gm_n in the shunt LDO case. Note that the impedance of any capacitor is $Z_C = \frac{1}{sC}$ where s is the complex domain frequency.

$$
\begin{aligned}
Z_{NFseries} &= \frac{1}{gm_pgm_{EA}(Z_{Cgdp}\|R_{OE}\|Z_{Cgsp}\|R_A)} \\
Z_{gmpseries} &= \frac{Z_{Cgdp}+R_{OE}\|Z_{Cgsp}\|R_A}{(gm_p-gm_{EA})(R_{OE}\|Z_{Cgsp}\|R_A)} \qquad (10)\\
Z_{NFshunt} &= \frac{1}{gm_ngm_{EA}(R_{OE}\|Z_{Cgdn}\|Z_{Cgsn})} \\
Z_{gmpshunt} &= \frac{Z_{Cgdp}+R_A\|R_B\|Z_{Cgsp}}{gm_p(R_A\|R_B\|Z_{Cgsp})} \\
Z_{gmnshunt} &= \frac{Z_{Cgdn}+R_{OE}\|Z_{Cgsn}}{(gm_n-gm_{EA})(R_{OE}\|Z_{Cgsn})} \qquad (11)
\end{aligned}
$$

The respective PSR using the short-circuit current i_{sc} and Z_{out} models for series and shunt LDOs are shown in Fig. 6(c) and (d).

2) PSR Pole/Zero for Series LDO: From Eq. 7, **poles occur when A_{IN} approaches infinity**, which happens when i_{sc} introduces infinite current or Z_{out} becomes open or infinite. **Zeros occur when A_{IN} approaches 0**, which happens when i_{sc} introduces zero current or Z_{out} becomes a short.

The series LDO's A_{IN} plot is shown in Fig. 6(i), with the pole and zero equations given in Table I. At low frequency, $i_{gmp} \approx 0$ because $R_A \gg R_{OE}$ and Z_{NF} dominates in Z_{out}, so $A_{IN} \approx i_{rop} \cdot Z_{NF} \approx \frac{1}{gm_{EA}R_{OE}gm_pr_{op}}$. A more exact expression, including R_L and r_{op}, is shown in Table I with the DC values. Table I first highlights the conditions for finding the poles and zeros in yellow, then shows the corresponding expressions for each pole and zero.

In this series LDO, **zeros are caused by zero i_{sc}**, while the **poles are caused by infinite impedance**. Note that infinite impedance is created by having parallel positive and negative impedances of the same magnitude. The poles and zeros for the series LDO are listed as follows and shown in Table I.

- **Zero 1**: The first zero occurs when $i_{rop}+i_{gmp}=0$ and solve for s. As frequency shifts toward the left-plane zero (LPZ), I_{gmp} becomes negative to oppose i_{rop} when the negative impedance of $Z_{cgdp}\|R_A$ starts to dominate the positive impedance of $Z_{cgsp}\|R_{OE}$.
- **Zero 2**: The second zero occurs when $i_{rop}+i_{gmp}+i_{Z_T}=0$. As frequency increases further, i_{gmp} becomes positive again due to Z_{cgsp} starting to short R_{OE}, but eventually settles to $gm_p \cdot C_{gdp}/(C_{gdp}+C_{gsp}) \approx \frac{1}{3.5\ gm_p}$, which may be smaller in magnitude than r_{op}. Finally, i_{Z_T} begins to oppose the other two currents; at a higher frequency, i_{Z_T} completely cancels the other two currents. (The calculated Zero 2 from the model occurs at a higher frequency than the simulated model, as some high-frequency effects are not captured in the equation.)
- **Complex Conjugate Poles (CCP)**: As an approximation, the CCP occurs when $Z_{NF}=-Z_{CL}$, i.e., when the negative feedback is no longer active around the UGB. Reducing LG or increasing C_L can reduce peaking.

Another pair of high-frequency poles and zeros for the series LDO are not plotted or included in the equations. An additional zero is caused by R_{ESR} limiting the current of C_L, and an additional pole is caused by Z_{CB} becoming smaller than R_{ESR}.

Analysis in Table I shows that Zero 1 and CCP 1 & 2 are highly correlated with similar terms, such that Zero 1 always precedes the CCP, introducing peaking with no apparent cancelation.

3) PSR Pole/Zero for Shunt LDO: The shunt LDO's A_{IN} plot is shown in Fig. 6(j), with the corresponding equations

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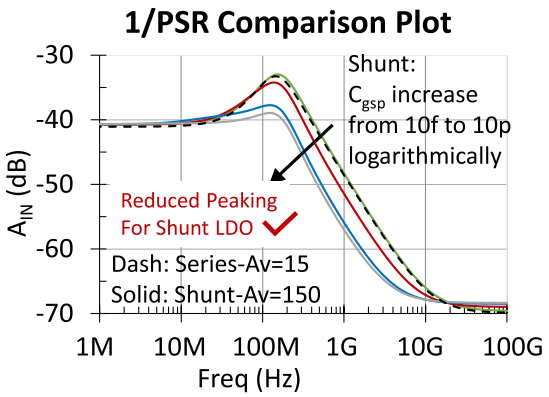


Fig. 7. A_{IN} response vs. frequency for a sample series LDO (dashed line) without increasing in Cgsp and shunt LDO (solid line) as the Cgsp increases from 10 fF to 10 pF. As Cgsp in shunt LDO increases.

listed in Table I. The zeros and poles of the series and shunt LDOs behave similarly, except that the shunt LDO introduces an extra zero and pole because it has three independent storage components: C_{gsn}, C_L, and C_{gdp}. The additional pole and zero are listed below:

- **Zero 3**: The additional zero occurs when Z_{NF} approaches zero.
- **Pole 3**: The additional pole occurs when $Z_{CL} = -Z_{gmp}$.

The extra pole and zero in the shunt LDO add another degree of freedom to the PSR response, which can help reduce peaking in the mid-to-high frequency range. As shown in Table I, Zero 1 and CCP no longer share the PMOS terms as compared with series LDO; Zero 1 consists of parasitics from the driving PMOS branch, whereas CCP originates from the shunt NMOS path. The additional Pole 3, on the other hand, shares similar terms with Zero 1. When $C_{gsp} \gg C_{gdp}$, Pole 3 and Zero 1 cancel, as also illustrated in Fig. 6(j). The effect of increasing C_{gsp} in the shunt path is shown in Fig. 7. As C_{gsp} increases, the PSR improvement in the mid-to-high frequency range is approximately 6 dB.

The desirable cancellation occurs between the initial Zero 1 and additional Pole 3, but would the additional Zero 3 cause peaking? As shown in Table I, PSR Zero 3 does not occur before the CCP and therefore does not cause peaking, due to its smaller denominator (R_{OE} C_{gn}) compared with the denominator of the real part of the CCP (2 R_{OE} C_{gn}).

C. Security Analysis

1) CDSA Model Description: CDSA models for series and shunt LDOs are shown in Fig. 6(e) and (f). Z_T represents the impedance connected from output node to the supply node and Z_B is the bottom impedance from output node to ground. CDSA model is very similar to the PSR model with just the impedances. Note that R_A is not AC-grounded through the supply node and is therefore not included in Z_T. In the model, Z_T includes a factor to exclude R_A as follows:

$$Z^* = Z \cdot \frac{R_A + R_B || Z_{Cgsp}}{R_A}. \tag{12}$$

CDSA uses the concept of current division and finds

$$A_L = i_{in}/i_l = Z_{out}/Z_T, \tag{13}$$

where Z_{out} is the same as in the PSR model but without R_L. From this equation, A_L poles occur when A_L approaches infinity (Z_{out} approaches infinity or Z_T approaches 0), and zeros occur when A_L approaches zero (Z_{out} approaches zero or Z_T approaches infinity). **The poles of A_L are the same as those of A_{IN}, while the zeros mostly occur when Z_T approaches infinity**.

2) CDSA Pole/Zero for Series LDO: The series LDO A_L plot is shown in Fig. 6(k) with the pole and zero equations in Table I. DC A_L is approximately $R_A/(R_A + R_B)$ and when $R_A \gg R_B$, A_L is around 0 dB as the load variation is directly supplied from the supply through the PMOS. The poles occur when Z_{out} approaches infinity, which are very similar to the ones in A_{IN} if Z_{CL} has reduced to less than R_L at the first pole frequency. The zeros occur when Z_T approaches infinity, at a frequency higher than the poles.

3) CDSA Pole/Zero for Shunt LDO: In the shunt LDO, the A_L plot is shown in Fig. 6(l) with the pole and zero equations in Table I. DC A_L is given by $1/(gm_{EA} \cdot R_B \cdot gm_N \cdot r_{op})$, which can be significantly better than the series case as changes in the load current are supplied by the shunt-path NMOS. The poles and zeros of the shunt A_L are very similar to the ones from the shunt A_{IN}, because Z_{out} is the same without R_L, and $1/Z_T$ is also very similar to gm_{sc} when $R_A \gg R_B$.

As shown in Fig. 6(k) and (l), with a similar gain-bandwidth product in both series and shunt LDOs (including an extra C_{gsp} = 10 pF in the shunt), the shunt LDO consistently has better A_L, except at high frequencies where the difference is negligible. As observed previously in the PSR analysis for the shunt LDO, C_{gsp} also improves the CDSA response in the mid-to-high frequency range. To understand the benefit of the shunt more intuitively, note that Z_{NF} has the lowest impedance from low frequency up to the UGB. In the series LDO, Z_{NF} is connected to the input supply, whereas in the shunt LDO, Z_{NF} is connected to ground. Since current flows through the lowest-impedance path, the series LDO exhibits worse security performance due to the low-impedance path being connected to the supply.

D. Large Signal PSR and CDSA Response

Previous sections analyzed small-signal responses for PSR and CDSA; however, digital circuits can introduce some large-signal effects. Fig. 8(a) shows the large-signal A_{IN} response versus the amplitude of a voltage sine wave at different frequencies. The sine wave is added in series with V_{IN} at 1.2 V. In general, PSR degrades as the amplitude increases, except for the 1 GHz case in the series LDO. For lower frequencies below the UGB (100 MHz), the series LDO exhibits better PSR compared with the shunt LDO. However, for mid-to-high frequencies above the UGB, the shunt LDO shows better PSR response due to the reduced peaking around the UGB. Additionally, at mid-to-high frequencies, the load capacitance is more effective, resulting in a smaller reduction in PSR as the amplitude increases.

Fig. 8(b) shows the large-signal A_L response versus the amplitude of a current sine wave at different frequencies, placed in parallel with I_L at 500 μA. Note that the total I_L is limited to be greater than 0 at all times. Reduction in CDSA is also observed as amplitude increases. As expected, for frequencies below the UGB, the series LDO has around 0 dB CDSA, indicating minimal security, while the shunt LDO provides better large-signal CDSA response,

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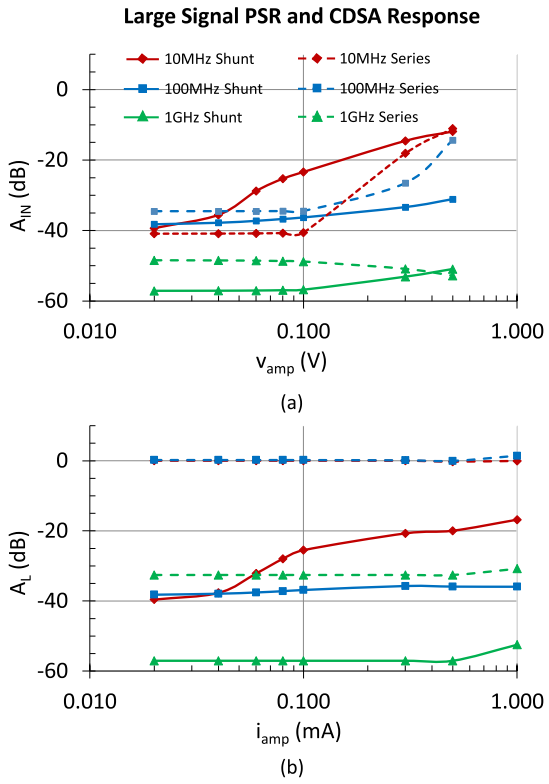


Fig. 8. (a) Large signal PSR response with v_{amp} in series with V_{IN} = 1.2 V. (b) Large signal CDSA response with i_{amp} in parallel with I_L = 500 μA, with total $I_L > 0$ A.

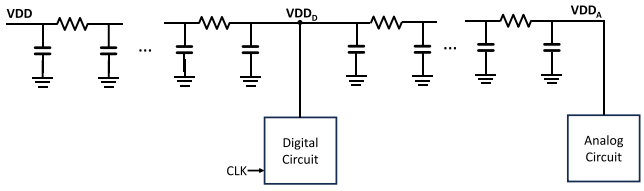


Fig. 9. Power grids modeled as RC distributed network from external pad to the digital VDD and analog VDD.

consistent with the small-signal analysis. At a high frequency of 1 GHz, C_L can compensate for a higher Δi_L due to the higher dv_{out}/dt ($i = C \cdot dv/dt$), so the series LDO also provides some security. Across the simulated frequency range, the shunt LDO consistently performs better in CDSA than the series LDO.

E. LSI Analysis

LSI is the inverse of the transfer function from the load current to the analog VDD (eq. 3). The LSI model uses an RC distributed network from the pad VDD to the digital VDD (VDD_D) and analog VDD (VDD_A), as shown in Fig. 9. Assuming that the distributed network has a wide bandwidth, the capacitors can be ignored, and the model can be simplified to Fig. 10. The transfer function becomes

$$A_{LSI} = \frac{v_{vdda}}{i_l} = A_L \cdot \frac{R_{DN1} \cdot R_{ANA}}{R_{DN1} + R_{DN2} + R_{ANA}} \tag{14}$$

From this transfer function, LSI (or A_{LSI}) is directly proportional to CDSA (or A_L), and A_{LSI} directly shares the poles and zeros of A_L.

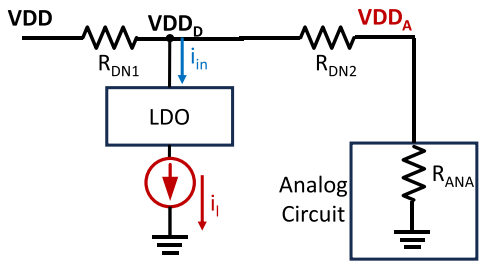


Fig. 10. Simplified LSI model with distributed network.

F. Analysis Summary

PSR for the series and shunt LDOs is analyzed using the $gm_{sc}Z_{out}$ method. CDSA is analyzed using current division, and LSI is proportional to CDSA. The key observations are summarized as follows.

For PSR:

- With similar LG and UGB, both series and shunt LDOs exhibit comparable low-frequency PSR proportional to LG.
- The shunt LDO introduces an additional pole and zero, and by utilizing the pole/zero cancellation with C_{gsp}, shunt LDO mitigates peaking around the UGB and improves mid- to high-frequency PSR by approximately 6 dB.
- Low-frequency large-signal PSR degrades with increasing input amplitude, whereas high-frequency signals benefit from load capacitance compensation, particularly in the shunt LDO.

For CDSA:

- The series LDO provides negligible CDSA benefit at low frequency, as load variations are directly supplied through M1 from the input.
- The shunt LDO CDSA mirrors the PSR characteristics, offering significantly higher benefit proportional to LG (around 40 dB).
- Large-signal CDSA shows similar trends as PSR: degradation occurs at low frequency with high amplitude, while high-frequency performance remains robust.

For LSI:

- Since LSI is proportional to CDSA, improvements in shunt CDSA directly enhance LSI, providing stronger isolation between digital load variations and the analog supply.
- Lower R_{ANA} and higher R_{DN2} can further improve LSI.

III. Design Considerations for Shunt LDO

From the previous discussions on the PSR and CDSA of the different LDOs, the advantages of the shunt LDO—improved PSR in the mid-frequency range and better CDSA across the frequency of interest—are clearly shown. However, there are additional design considerations associated with the shunt LDO, as illustrated in Fig. 11.

The first design consideration is the current efficiency of different types of LDOs, shown in Fig. 11(a). Two factors contribute to the lower efficiency in the shunt LDO: 1. The extra shunt path causes the shunt LDO to consume an additional 2 ~ 10% of the load current. 2. To achieve a similar loop gain, the shunt LDO requires an extra gain stage compared with the series LDO. Note that for LDOs designed for smaller loads, the PMOS gate capacitance is not necessarily large; therefore,

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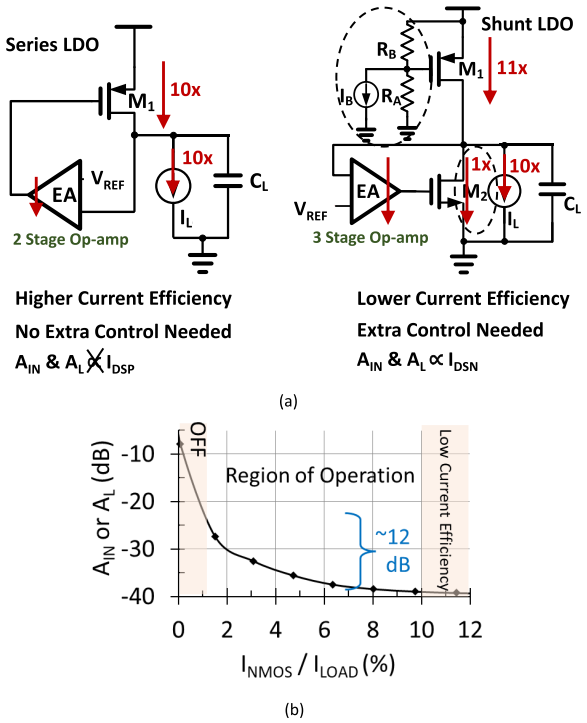


Fig. 11. (a) Extra design considerations for shunt LDO compared with series LDO. (b) DC A_{IN}/A_L vs. ratio of the shunt NMOS current and load current.

the EA design is constrained by the minimum transistor size rather than the slew rate. Consequently, an extra stage in the EA driving the shunt NMOS draws more current than the EA driving a larger PMOS.

Another consideration for the shunt LDO is the need to control both the driving PMOS and the shunt NMOS, which requires more complex logic, as shown in Fig. 11(a). As discussed previously, the shunt NMOS is controlled by the feedback EA loop. The driving PMOS in the shunt LDO can be controlled using either analog [29] or digital [23] methods. To achieve the security benefit, the shunt LDO requires a faster feedback response to the shunt NMOS than to the PMOS, and it should maximize utilization of the shunt path—conditions that may not be fully met with analog control [29]. A more digital SMC is therefore more appropriate, as the PMOS control can be programmed to be delayed, as in [32]. The drawbacks of a traditional digital LDO are discussed in the next subsection, Sec. III-A.

Another consideration is that the shunt LDO's loop gain is heavily dependent on the current through the NMOS, or more specifically, the g_m of the NMOS. This dependence on I_{DSN} arises because the gain can be significantly reduced, as shown in the DC equations for the shunt LDO in Table I. As illustrated in Fig. 11(b), A_{IN} and A_L improve with increasing current in the shunt path, with the improvement saturating around 8% to 10%. This implies that the PMOS control must be more precise to ensure the shunt path consistently draws approximately 10% of the load current. Therefore, digital control of the PMOS may be easier to implement. In series LDOs, the PMOS stage gain is more consistent while operating in saturation, making it less dependent on the current.

For high-frequency performance, the sizing and placement of the load capacitor and supply decoupling capacitor become more important than the shunt feedback path. The shunt path only improves the response up to the UGB, assuming minimal peaking. Beyond the UGB, the load capacitor becomes the main regulator.

A. Traditional Digital LDOs in PSR and CDSA

Due to reductions in technology nodes and lower supply voltages, digital LDOs have gained traction for providing smaller dropout voltages and faster large-signal transient responses compared with analog LDOs. However, the switching nature of the pass transistor, which has a low r_{op} in digital LDOs, can significantly degrade PSR and CDSA performance.

Low dropout voltage for higher power efficiency is not the primary concern in the field of security, as many security measures require several times the load power [38]. The low r_{op} in digital LDOs, which leads to inferior PSR and CDSA performance, is a more critical concern. Therefore, biasing the driving PMOS in the saturation region with digital control is necessary to achieve higher r_{op} and to provide a delayed feedback to the driving PMOS relative to the shunt NMOS.

B. Shunt LDO Considerations to Accommodate Different Loads

The analysis in Sec. II only considered the condition where I_L = 500 μA. To extend the shunt LDO operation to either lower or higher load limits, additional considerations must be taken into account. For both extremes, it is important to have a controllable number of slices for both the driving PMOS and the shunt path, as well as to maintain approximately 10% of the load current through the shunt path. The lower end of the load current is constrained by the desired current efficiency, which is limited by the EA assuming other control logic consumes much less power than the EA.

At the higher end of the load current, many undesirable phenomena start to occur, such as the PMOS entering the linear region, reduction of LG, ineffectiveness of the buffer, and decreased phase margin. Therefore, several factors need to be considered during the design stage: 1. sizing and number of the driving PMOS devices to maintain a high r_{op} for low-frequency PSR and CDSA; 2. achieving the desired LG under high load conditions, which also affects low-frequency PSR and CDSA; 3. buffer sizing for faster transient response; 4. larger C_L to improve phase margin. Note that depending on the sizing, capacitor choice, and layout of C_L, R_{ESR} may no longer be negligible, as it can introduce an additional zero and degrade the phase margin.

IV. Existing Shunt LDO Results and Comparison

Fig. 12(a) shows a sample shunt LDO design that can be paired with different current sources (CS) in Fig. 12(b)-(d) and different LNFB configurations in Fig. 12(e)-(f) [8], [21], [22], [23], [24], [25].

As indicated by the DC equations in Table I, increasing the shunt LDO's PSR and CDSA can be achieved by increasing either r_{op} or the LG. Increasing r_{op} is accomplished through the different current sources in Fig. 12(a)-(c); this increases the dropout voltage and reduces efficiency, but these are not the primary concerns of this paper, as the focus is on PSR and CDSA. The LG is influenced by the LNFB in Fig. 12(e)-(f).

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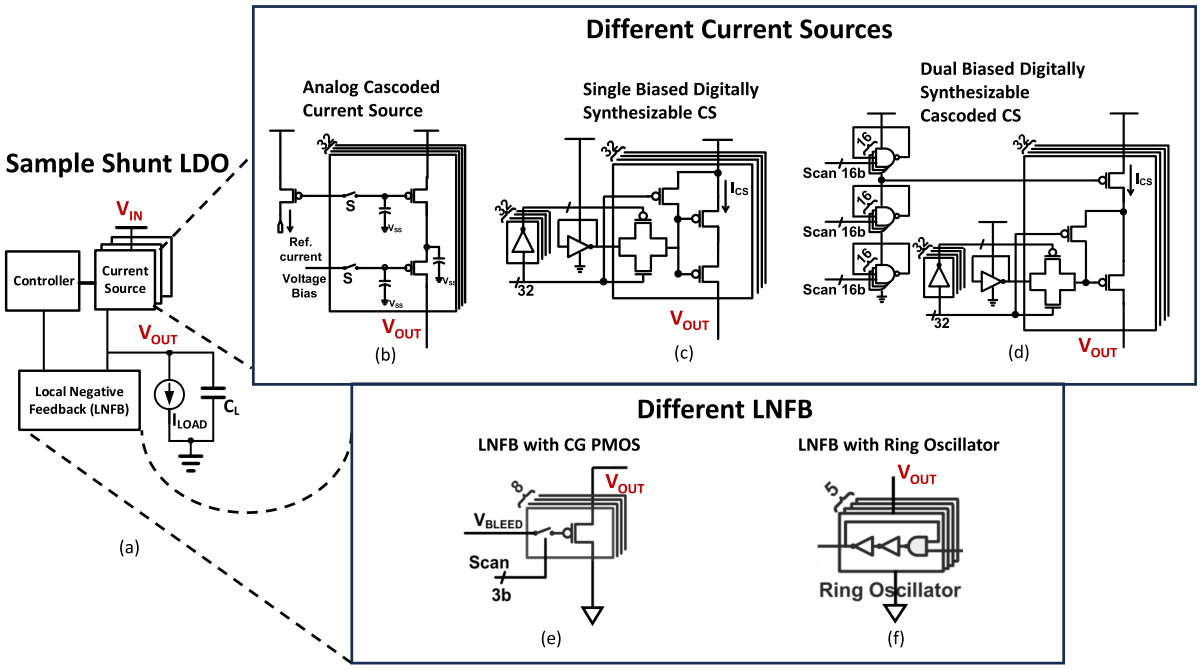


Fig. 12. (a) Sample Shunt LDO with different existing current sources and local negative feedback (LNFB). (b) Analog cascoded current source (CS) with dual analog biasing [21], [22]. (c) Synthesis-friendly CS with single biasing [8], [23]. (d) Synthesis-friendly cascoded CS with dual biasing [24], [25]. (e) LNFB with a common gate PMOS with an analog bias [21], [22]. (f) LNFB with ring oscillator [8], [23], [24], [25].

TABLE II
DIFFERENT LDO ARCHITECTURES CDSA COMPARISON

	#	Configuration	DC CDSA Equation	Condition	CDSA DC Calculated (dB)	CDSA @100 MHz Simulated (dB)	Comment
	0	Analog Series LDO Fig.6(a)	$(R_A + R_{OE})/R_A$	$gm_{EAP} = A_{EAseries} * gm_P = 15 * 6.5mS$ $r_{op} = 1\,k\Omega$ $R_A \gg R_{OE}$	0	0	Security is low, due to the load current changes are supported through the supply.
Shunt LDO	1	Analog EA with common source NMOS Shunt Path with a Driving PMOS Fig.6(b)	$1 + gm_{EAN} r_{op}$	$gm_{EAN} = A_{EAshunt} * gm_n = 150 * 650\mu S$	40	39	Another gain stage is required for EA to achieve similar PSR as series LDO. CS NMOS needs to consume about 10% of load current.
Shunt LDO	2*	Analog Cascoded Driving PMOS Fig.12(b)	$1 + gm_{EAN} r_{opcasA}$	$r_{opcasA} = r_{opd}(1 + A_{casA})$ $A_{casA} = gm_d r_{opt} = 1\sim10$ $r_{opt} = r_{opd} = r_{op}$	46~60	-----	Cascoded PMOS increases top impedance, but will increase dropout voltage.
Shunt LDO	3*	Single Biased Digitally Synthesizable Driving PMOS Fig.12(c)	$1 + gm_{EAN} r_{opcasD1}$	$r_{opcasD1} = r_{opd}(1 + A_{casD1})$ $A_{casD1} = gm_d r_{opt} = 1\sim5$ $r_{opt} < r_{opd} = r_{op}$	46~55	-----	Same gate bias and same size PMOS. The top PMOS will be in linear region while bottom PMOS in saturation, thus ropt<ropd and $A_{casD} < A_{casA}$
Shunt LDO	4*	Dual Biased Digitally Synthesizable Cascoded Driving PMOS Fig.12(d)	$1 + gm_{EAN} r_{opcasD2}$	$r_{opcasD2} = r_{opd}(1 + A_{casD2})$ $A_{casD2} = gm_d r_{opt} = 1\sim10$ $r_{opt} = r_{opd} = r_{op}$	46~60	----	Similar to analog cascoded PMOS but digitally synthesizable.
Shunt LDO	5+	Analog Biased CG PMOS Shunt Path Fig.12(e)	$1 + gmp_{CG} r_{op}$	$gmp_{CG} = gm_n$ $r_{op} << r_{opCG}$	4	----	The loop gain is significantly reduced, thus much lower PSR and security
Shunt LDO	6+	RO Shunt Path Fig.12(f)	$1 + gmp_{INV} r_{op}/2$	$gmp_{INV} = \frac{1}{5} gm_n$ $r_{opINV} = r_{onINV}$	0.5	----	Due to the size of the inverter, inverter gm can be much less than CS NMOS or CG PMOS.
Shunt LDO	7	2 & 5 Fig.2(b)	$1 + gmp_{CG} r_{opcasA}$	--	7~18	49.2	ISSCC (2020) JSSC (2021)
Shunt LDO	8	3 & 6 Fig.2(c)	$1 + gmp_{INV} r_{opcasD1}/2$	--	1 ~ 3	34.1	ISSCC (2021) JSSC (2022)
Shunt LDO	9	4 & 6 Fig. 2(d)	$1 + gmp_{INV} r_{opcasD2}/2$	--	1~5	49.8	CICC (2022) OJ-SSCS (2025)

*: Assumes LNFB with an analog EA with common source NMOS shunt path

+: Assumes a single driving PMOS for current source

$A_{EASeries}$: error amplifier gain in the series LDO feedback loop
gm_p: driving pmos transconductance
r_{op}: driving pmos output resistance
R_A & R_{OE}: output impedance at the gate of the driving pmos
$A_{EAShunt}$: error amplifier gain in the shunt LDO feedback loop
gm_n: shunt path common source nmos transconductance
r_{opcas}: cascoded driving pmos equivalent output resistance
r_{opt} & r_{opd}: output resistance of the top & bottom of the cascode pmos
gm_d: bottom pmos transconductance in the cascoded driving pmos
gmp_{CG}: transconductance of the shunt path common gate pmos
gmp_{INV}: transconductance of the pmos in the RO's inverter
r_{opINV} & r_{onINV}: output resistance of the RO's inverter

Increasing the LG can degrade stability and may require additional compensation, which further reduces bandwidth and consequently decreases PSR and CDSA in the mid-frequency range.

A summary of the DC CDSA equations, simulated CDSA at 100 MHz, and associated values under different conditions for various security-focused shunt LDOs is shown in Table II. Note that the sample analysis in Sec. II assumes a high loop

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gain, so the constant terms in the DC CDSA equations were ignored; however, in this section, some loop gains may be small, so the constant terms are included. Frequency-domain analysis for the different security shunt LDOs is discussed in Sec. IV-C.

A. Effects of Different Current Sources

A cascoded design has been used for PSR improvement in series LDOs [27], [39] and for security improvement in shunt LDOs [22], [23], [24]. Fig. 12(b) shows a dual analog-biased cascoded PMOS with digital control over the number of current slices [21], [22]. The cascoded PMOS increases the original r_{op} to $r_{opd} \cdot (1 + gm_d \cdot r_{opt})$, where r_{opd} is the lower PMOS's r_{op}, r_{opt} is the top PMOS's r_{op}, and gm_d is the lower PMOS's transconductance. If $gm_d \cdot r_{opt}$ ranges from 1 to 10, the CDSA of the shunt LDO can improve by 6–20 dB.

Fig. 12(c) shows a stacked PMOS with a single digitally synthesizable bias [23]. When the stacked PMOS slice is on, the self-biased inverter sets both PMOS devices to VDD/2. With both PMOS devices being the same size and having the same gate bias, operating them in the above-threshold saturation region while maintaining a reasonable dropout voltage is unlikely. However, it is possible to operate the bottom PMOS in saturation while the top PMOS is in the linear region with initial tuning of VDD. In this case, r_{op} becomes $r_{opd} \cdot (1 + gm_d \cdot r_{opt})$, where $r_{opd} > r_{opt}$. The digitally biased cascoded PMOS will have a smaller $gm_d \cdot r_{opt}$ with the top PMOS operating in the linear region. Assuming $gm_d \cdot r_{opt}$ ranges from 1 to 5, the CDSA improvement is about 6–15 dB.

To further improve the design in Fig. 12(c), Fig. 12(d) shows a cascoded PMOS with dual digitally synthesizable biases. By turning the NAND gate on and off at different levels, the bias of the top PMOS can be tuned for optimal performance. The dual digital bias brings the CDSA response of the shunt LDO to levels similar to those achieved with dual analog bias.

B. Effects of Different Local Negative Feedback

To simplify the circuit for a more digitally friendly design and reduce power consumption from a high-gain shunt path, [21], [22] reduce the shunt analog feedback loop to a CG PMOS, as shown in Fig. 12(e). The CG PMOS provides the necessary DC current drain from the SMC; however, it reduces the overall loop gain, which in turn reduces the low-frequency CDSA of the shunt LDO. The small-signal resistance of the CG PMOS becomes approximately $1/gm_{CG}$, where gm_{CG} is the transconductance of the common-gate PMOS. Note that since the shunt path should draw less than 10% of the load current, gm_{CG} will be small. Since r_{op} will also be small compared with rop_{CG}, the overall DC rejection would only be around a few dB. Incorporating both techniques—a cascoded PMOS and a CG shunt PMOS—and assuming gm_{CG} is the same as gm_N, the total DC rejection for CDSA can range from 7 dB to 18 dB, which is still better than the series LDO at 0 dB for security.

Fig. 12(f) shows a ring oscillator (RO) as LNFB [23], [24]. Due to the switching nature of the RO, it provides more dynamic current changes than the previous analog methods. Fig. 13(a) shows the transient response of the inverter outputs in a RO. As shown, the rising and falling edges happen in sequential order, meaning that one inverter output rising is followed by another inverter output falling, while the rest

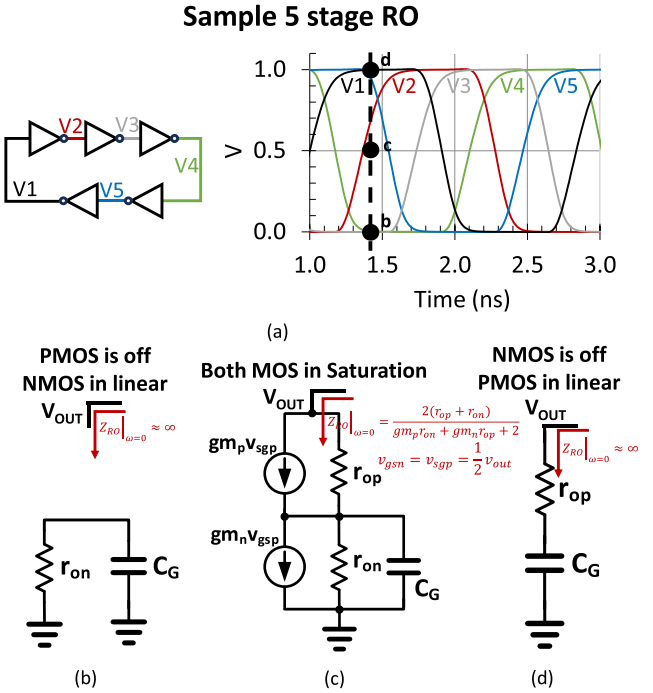


Fig. 13. Breakdown of how RO performs as a LNFB. (a) Output voltages for different inverters in the RO vs. time. (b) Point b of the inverter schematic when output is low (No LNFB). (c) Point c of the inverter schematic when output is rising or falling (Most LNFB). (d) Point d of the inverter schematic when output is high (little to negligible LNFB).

of the inverter outputs are either at supply or ground. This characteristic helps to explain how the RO contributes to security, as shown in Fig. 13(b)-(d).

Note that LNFB is used to drain the excess current while providing some negative feedback by providing a low-impedance path to ground (Z_{RO}); therefore, for analysis purposes, comparisons with different Z_{RO} are essential. 1. For the inverters with output grounded in Fig. 13(b), the PMOS is off while the NMOS is in the linear region, so any changes in the RO supply do not affect the currents through this set of inverters. 2. For the inverters with output at supply in Fig. 13(d), the PMOS is in the linear region while the NMOS is off. The r_{op} and the parasitic capacitance (C_G) of the next stage form a relatively high-impedance path at low frequencies; since C_G is low, these paths contribute minimally to security. 3. For the inverter that is at either rising or falling edges in Fig. 13(c), both PMOS and NMOS are turned on, forming a more resistive path to ground, which helps with security at lower frequencies. Note that the propagation from the output to the gate node of the inverter does not significantly change Z_{RO} because the current reduced in the PMOS-on stages is gained in the NMOS-on stages. Most benefit to security occurs when both PMOS and NMOS are in saturation due to the higher loop gain.

For an approximation, assume that at any given time, there is one inverter with both PMOS and NMOS in the saturation region and biased at $v_g = v_{out}/2$. Then assume $gmp_{INV} \approx gmn_{INV}$ and $rop_{INV} \approx ron_{INV}$; the resistance of this branch becomes $R_{RO} = (2rop_{INV})/(gmp_{INV} \cdot rop_{INV} + 1)$. The benefit of the RO in terms of PSR is diminished because, as R_{RO} is expected to be much larger than R_L, a low-resistance path to ground is not formed, and the PSR is reduced. The security at DC becomes $1 + gmp_{INV} \cdot rop/2$. Note that since

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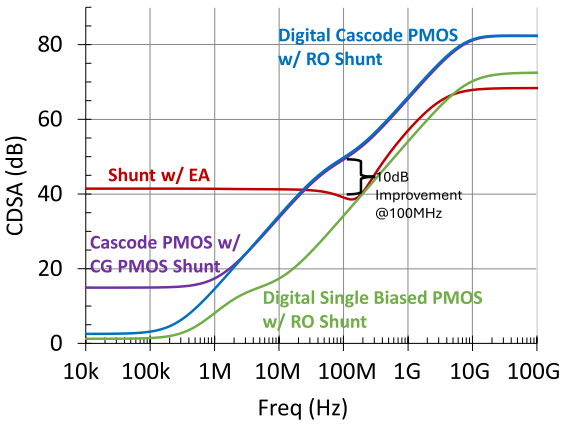


Fig. 14. Simulated CDSA frequency analysis for different shunt LDO architectures.

the inverters cannot be sized as large as the PMOS in the CG case to consume more current, gmp_{INV} can be much smaller than gmp_{CG}, while r_{op} stays relatively similar due to the constraint from the driving PMOS; thus, the security can be less than 1 dB.

C. CDSA Frequency Analysis

The CDSA frequency analysis for different shunt LDO architectures is shown in Fig. 14. Fig. 14 utilizes the models mentioned in Sec. IV-B. The simulation assumes a 500 μA load current, with the shunt path draining approximately 50 μA. The low-frequency CDSA matches the calculated values. The analog shunt LDO with an EA maintains high CDSA up to the UGB, whereas the other architectures that utilize different DC current drain paths without high-gain feedback trade off low-frequency CDSA for high-frequency CDSA. For the architectures with a DC current drain path, the low-gain feedback contributes to the low-frequency CDSA response; however, the low UGB allows the CDSA to start improving from a lower frequency. Even though UGB is the term used to conform with the previous analysis, the point CDSA starts to improve is when Z_{CL} starts to shunt the drain path. As a result, the cascode PMOS with CG PMOS shunt path and the digital cascode PMOS with RO shunt path show a 10 dB CDSA improvement compared with the shunt LDO with EA. The digitally single-biased PMOS with RO shunt path has a worse response due to the coupled PMOS contributing to a higher CS capacitance, hence lowering Z_T and reducing CDSA.

To verify the RO model discussed in Sec. IV-B, a harmonic balance AC simulation was run on a 41-stage RO shunt path [8], [23] and compared with the RO model. Similarity can be observed between the two methods, which verifies the validity of the RO model.

D. Comparison Table

Table II summarizes the existing shunt LDO architectures with the CDSA equations, along with the respective calculated numbers and simulated results. Unless specified, the different configurations assume either a single driving PMOS as the current source or an analog EA with a common source NMOS in the shunt path.

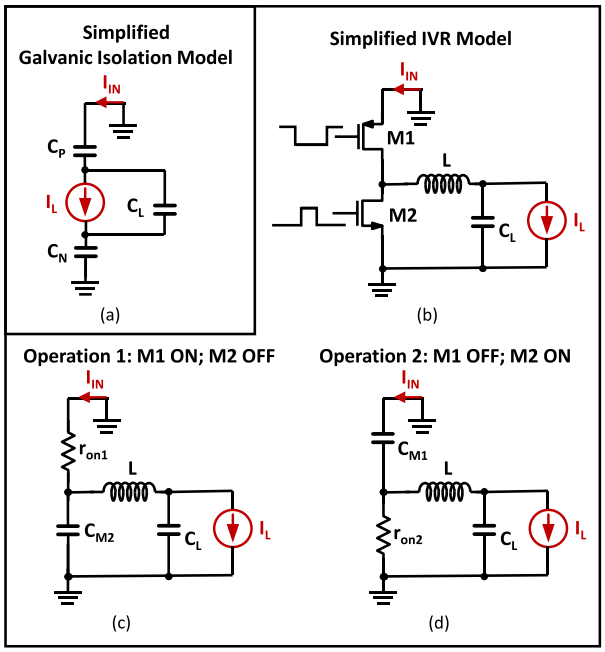


Fig. 15. Simplified CDSA models for other isolation architectures. (a) Simplified galvanic isolation model. (b) Simplified IVR model with non-overlapping clocks. (c) IVR model operation 1 when M1 is on and M2 is off. (d) IVR model operation 2 when M1 is off and M2 is on.

As a summary, a cascoded PMOS stage improves CDSA, PSR, and LSI at the expense of higher dropout voltage and lower efficiency. A high-gain and high-BW feedback path is also important to improve these three metrics. If the focus is specifically on CDSA rejection around the clock transition, low-frequency CDSA can be traded off with a low-UGB shunt path to achieve improved high-frequency CDSA. The last stage of the feedback path to V_{OUT} should not be a common-source stage due to the comparatively low output resistance contributed by the driving PMOS, along with the inherently low gm from the limiting shunt path current; this gain can be well below unity, negating the benefits of a gain stage. Following inspiration from the flipped voltage follower LDO, the last stage of the feedback path is best implemented as a source-follower-like stage to counterbalance the low output resistance from the driving PMOS; this gain can be very close to unity. For higher-frequency responses above the UGB, the sizes of C_L and decoupling capacitance become important—the higher the C_L and decoupling capacitance, the better the response.

V. Existing Isolation Architecture Analysis

The galvanic isolation architecture in Fig. 3(a) can be simplified into Fig. 15(a), where the off transistors are modeled as C_P and C_N. A current divider can now be used to find

$$A_L = \frac{Z_L}{Z_P + Z_N + Z_L}. \tag{15}$$

This equation shows that, to improve security, Z_P and Z_N should be maximized while minimizing Z_L.

The digital LDO in Fig. 3(b) can be thought of as a digital series LDO, where the isolation is severely diminished by the low r_{op}, as discussed in Sec. III-A.

The IVR in Fig. 3(c) can be simplified into Fig. 15(b), where the digital controller is not shown.

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Note that non-overlapping clocks are used to control M1 and M2. Two operation modes for the IVR are now shown in Fig. 15(c) and (d). During operation 1, at low frequency, Z_L is minimum and r_{on1} is also small because M1 is completely turned on. Operation 1 is very similar to a typical series digital LDO, where security is diminished. Operation 2, on the other hand, can provide extra security. Intuitively, I_L is first compensated by C_L, next isolated by L, especially at high frequencies, and lastly drained through r_{on2}. At low frequencies, I_L will be drained through r_{on2} with essentially no leakage to the supply. At high frequencies, A_L can be calculated using a series of current dividers:

$$A_L = \frac{(Z_L + Z_{C_{M1}} \| r_{on2}) \| Z_{C_L} \cdot (Z_{C_{M1}} \| r_{on2})}{(Z_L + Z_{C_{M1}} \| r_{on2}) \cdot Z_{C_{M1}}}. \tag{16}$$

The architectures in Fig. 3(b) and (c) only provide limited security; therefore, both approaches require additional noise injection to improve security.

VI. Conclusion

As the need for better isolation of sensitive circuitry and more physically secure devices grows, this work provides a comprehensive analysis of series and shunt LDO architectures with respect to PSR, CDSA, and LSI. Using both small-signal and large-signal models, we demonstrate that shunt LDOs offer superior CDSA and LSI performance and mitigate mid-frequency PSR peaking through an additional pole-zero pair. Key design parameters—including shunt NMOS sizing, driving PMOS biasing, loop gain, and load capacitance—are shown to critically affect these metrics. We analyze various existing shunt LDO architectures, highlighting trade-offs between DC rejection and high-frequency CDSA response. Frequency-domain simulations validate the models and quantify improvements across architectures. Alternative isolation countermeasures, such as galvanic isolation, digital LDOs, and IVRs, are also examined. Isolation methods are most effective when used in conjunction with noise injection or logic-level current masking to achieve optimal SCA results.

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